

P18: Robot Navigation using 3D LiDAR Maps

Motivation and Vision

3D information significantly enriches robot capabilities, particularly for tasks like obstacle avoidance, mapping, and planning in complex environments. Furthermore, 3D LiDAR sensors are becoming increasingly accessible and affordable, e.g., the Robosense Airy or Hesai JT128. The first step in exploiting 3D LiDARs for robot navigation is to use them for state estimation. Reliable odometry and SLAM systems for 3D LiDAR sensors exist, enabling robots to effectively explore and map unknown environments. However, there is, to some degree, a lack of reliable, usable 3D localization systems that allow a robot to localize itself within prebuilt maps. For example, in ROS context, most open-source solutions remain 2D-based.

This project aims to bridge this gap. A ROS2 navigation system will be developed which allows a robot to move autonomously from point A to point B, but with the core focus on using 3D LiDARs to perform 3D localization in a prior map. This will extend the localization part of the classic ROS navigation stack into 3D.. While the project will initially start its development in simulation, the goal of the project is to demonstrate this navigation system running real-time on a physical robot.

Important Note: This project will require significant engineering effort to realize its full potential, and software development for this project must be implemented in C++.

Interpreted languages are not suitable for core components, as the aim is to demonstrate this approach working in real-time on a real robot. Students who wish to take this project are strongly advised to have attended or at least attend in parallel any of the C++ courses offered in their course of study.

Project Objectives

The main objective is to design and implement a robust navigation system using 3D localization operating at sensor framerate. As a minimal demonstration, this system must be usable indoors; outdoor operation is also desirable. The project will initially start out being implemented in simulation to ensure rapid progress, with the goal to run on a real robot towards the end of the master project.

Specifically, the students will:

- Learn ROS 2 and utilize it as the core middleware stack for all subsequent tasks.
- Implement a 2D navigation stack to familiarize themselves with core robotics concepts and gain practical hands-on experience.
- Implement a 3D LiDAR mapping system to generate high-quality maps suitable for localization.
- Implement a 3D localization system that can operate with or without an initial guess, exploring global registration approaches.
- Finally, demonstrate autonomous start-to-end goal navigation using a real robot.

Tools and Hardware

The project will initially start out in simulation, specifically using ROS 2 and Gazebo to model the robot and sensor inputs. This allows for rigorous testing of algorithms without hardware constraints. As the project progresses and depending on the system's maturity, deployment on real robots will be investigated to validate the approach in real-world conditions. The primary language of implementation will be C++.

Key Technical Components to be Developed

The project is divided into several distinct technical components. Students can distribute responsibilities among team members, but they must ensure tight system integration for the final demonstration.

2D Navigation Stack

Students will first develop a visual marker-based pose estimation system. This will initially serve to build 2D maps of an environment and later as a reference pose tracking system for evaluating the accuracy of the final 3D localization. This pose estimation system can be used to integrate range scanner measurements into a 2D (occupancy) map. Once a map is available, a classic scan-to-map registration-based 2D localization system will be implemented. The components implemented here will serve as the building blocks for the project goal of a 3D localization system. To achieve autonomous robot point-goal navigation, the students will also implement a planner and controller, effectively creating a full ROS 2 navigation stack.

3D Mapping System

Using the visual marker-based pose estimation from before, a 3D occupancy mapping system will be implemented. The final map will be filtered for dynamics estimated from the probabilistic mapping process. The resultant map will be used in the project's aim of a localization system.

Scan-to-Map Registration Module

The localization will primarily rely on an ICP-based scan-to-map registration module to enable pose tracking in a prior map. To maintain sensor frame-rate in pose tracking, the students will investigate efficient data structures to represent the map, and implement an efficient algorithm to compute the necessary data associations for scan registration. The students will also investigate suitable motion models to serve as the initial guess for the ICP registration process.

Initial global registration

Scan-to-map registration generally requires a good initial guess of where the robot is in the environment. While in operation, a motion model can be used to provide an initial guess for subsequent scans; however, when the robot is first turned on, this guess is not immediately available. To solve this, the students will investigate different global registration approaches, and implement or adapt the necessary components therein (e.g., a local-map approach

requires odometry). The initial guess from the global registration can be refined through fine geometric alignment. If required, simplifying assumptions on robot motion can be made, such as requiring the robot to stay still during the global initialization phase. Once this module localizes the robot in the prior map, the scan-to-map registration module can take over to maintain pose tracking. Failure cases of the global registration will be handled through hard system resets, as maintaining multiple pose hypotheses is beyond the scope of this project (though it can be an extension point for the motivated group of students).